TopSo'Linux: TopSolid on Linux

Short guide to installing and using TopSolid 7.20 with its local PDM on Linux

TopSo'Linux is a patched Wine plus an installer that sets up TopSolid from TopSolid's installation media, your own copy or downloaded from TopSolid's server: TopSolid, SQL Server Express with the local PDM server, the Sentinel license server and TopSolid'Update. No Windows needed.

TopSo'Linux is unofficial and not affiliated with or supported by TOPSOLID SAS.

1. What you need

- Ubuntu 24.04 or newer, or another 64-bit distribution with glibc 2.38 or newer.
- The TopSolid installation media (the folder with `Setup.exe`), or let the installer download them from TopSolid's server (19 GB).
- About 45 GB of free disk space (65 GB when downloading) and an internet connection.
- A GPU with a working OpenGL driver. Laptops with NVIDIA and Intel/AMD graphics use the integrated GPU; right-click TopSolid in the app menu and choose *Launch using Discrete Graphics Card* for the NVIDIA one.

2. Installing

- Download the newest `TopSoLinux-Installer-<version>-x86_64.AppImage` from the releases on GitHub, make it executable (file properties, or `chmod +x`) and start it.
- Choose the installation media (found automatically in Downloads, your home folder or a USB drive), another folder, or the download from TopSolid's server.
- The installer puts TopSolid in your app menu with **Gear Lever** and installs SpaceMouse support (spacenavd); installing them asks for your password.
- TopSolid's own installer opens: choose the modules you use, click Install and accept the license agreement. **Its installation of SQL Server stops with an error. That is expected:** TopSo'Linux stops it and installs SQL Server itself. Close the errors and the setup.
- Wait: the whole installation takes 30 to 60 minutes. If it is interrupted, start the installer again and it continues where it stopped.

Everything is installed in `~/.local/share/topsolinux`. The result is `TopSolid-7.20-TopSoLinux-<version>-x86_64.AppImage`, a complete, self-contained TopSolid (Gear Lever renames it to `topsolid_7.20_topsolinux-<version>.appimage`).

3. The TopSolid AppImage

- Right-click TopSolid's icon in the app menu for *TopSolid'Update*, the license tool, *SpaceMouse settings*, *Wine settings*, *Stop background services* and *Check for TopSo'Linux updates*, which updates the AppImage to the newest TopSo'Linux. TopSolid also looks for one once a day when it starts.
- *Report a problem* makes a zip in your Downloads folder with the logs and what the computer has, to send to whoever helps you. It can record TopSolid or TopSolid'Update while the problem happens.
- Another computer: copy the AppImage there and start it, and choose *Install* (app menu, with Gear Lever) or *Launch*. The first start sets TopSolid up in `~/.local/share/topsolinux`, which takes a few minutes. Each computer needs its own license.
- Only use it on computers you are licensed for, and do not publish it: it contains TopSolid, SQL Server Express and Microsoft's runtimes.

4. First start

- License: as on Windows. A fixed license on a Sentinel USB protection key works too: on the first start with the key plugged in, TopSo'Linux offers to install Thales' Sentinel runtime (asks for your password once).
- PDM connection: choose **Local PDM Server** and leave user and password empty.
- Libraries: import the ones you need from TopSolid.
- New Project: enter a name, choose Blank Template and click the green check mark.

SQL Server, the local PDM and the license server start by themselves and stop again when you close TopSolid. The very first start of the PDM can take a few minutes; TopSo'Linux shows what it is doing until TopSolid shows its own splash screen. *Command Prediction* is off in a new installation (it keeps a processor core and several GB busy); turn it on in TopSolid's options if you want it.

5. Updates

Service packs come through TopSolid'Update (right-click the icon). They update the installation in `~/.local/share/topsolinux`, not the AppImage file, so Gear Lever's update button does not apply. After *Download now* the window disappears: the download runs in the background, and a little later TopSolid asks to install it.

6. Useful commands and tips

The AppImage also accepts a command, e.g. `TopSolid-7.20-TopSoLinux-<version>-x86_64.AppImage scale 150`:

`scale 150`	text and icons at 150% (100 = normal); by default TopSolid follows the desktop
`scale auto`	follow the desktop's scale again
`update`	start TopSolid'Update
`license FILE`	install a license file
`license-tool`	TopSolid's license tool
`spacemouse`	SpaceMouse speeds and axes
`stop`	stop SQL Server, the PDM and the license server
`topsolinux-update`	update this AppImage to the newest TopSo'Linux, if there is one
`update-check off`	don't look for TopSo'Linux updates at start (*on*: once a day again, the default)
`debug`	TopSolid hangs or typing goes nowhere: write its windows, keyboard focus and stacks to a file in your home folder, for a bug report
`report`	a zip with the logs for a problem report (same as *Report a problem*)
`record`	start TopSolid (*record update*: TopSolid'Update) with timestamps in the log; closing it makes the report
`winecfg, regedit, shell`	Wine tools, for troubleshooting

- Resize a window: hold Super (the Windows key) and drag with the middle mouse button, or press Alt+F8. Dragging the window edge does not work.
- The close button does not work in some dialogs: use the green check mark or the red cross at the bottom, or Esc.
- SpaceMouse: works through spacenavd, no 3Dconnexion driver needed. Speeds and axis directions: right-click TopSolid's icon, *SpaceMouse settings*. Changes work right away.
- Drive Z: is your home folder, Y: the rest of the system (USB sticks, other drives). Images, PDFs and text files that TopSolid opens, such as a saved screenshot, open in the desktop's own programs.

- Numbers and units follow the desktop's *Formats*: Dutch formats give a decimal comma and mm. To change them: `regedit`, `HKCU\Control Panel\International` (`sDecimal`, `iMeasure` 0 = metric).
- An environment shows as background only with a *Perspective* or *Conical* camera, with *Environment as Background* on (in the render mode menu, per document). With a parallel camera (Isometric, Front, ...) the background is one plain colour.
- The OpenCL device list in the display options stays empty: Wine has no OpenCL that shares TopSolid's OpenGL. Realistic rendering does not need it.
- Ubuntu: do not install the package **fuse**, it removes parts of the desktop. What AppImages need is already there.

7. If something goes wrong

- Logs: `~/.local/share/topsolinux/install.log` (installation) and `topsolid.log` (TopSolid) in the same folder.
- *Unable to connect to the PDM server* right after starting: wait a minute, the PDM is still starting. If it stays, run the `stop` command and start again.
- Do not run two TopSolid installations at the same time: they use the same ports.
- Backup: copy `~/.local/share/topsolinux` while TopSolid is closed. Your PDM projects are in it. Or package your projects in TopSolid.
- Newer TopSo'Linux: start the new installer and choose *Update the AppImage*. Your installation and projects stay. Starting over: choose *Replace*; your PDM projects are lost then. Both put older TopSolid AppImages in the trash and remove their app menu entries: they would start the installation with their old Wine.

How this was made

Completely vibe coded: the Wine changes, the installer and this guide were written by Claude Opus 5.5 (Anthropic), directed and tested by a TopSolid user who wanted TopSolid on a Linux desktop.